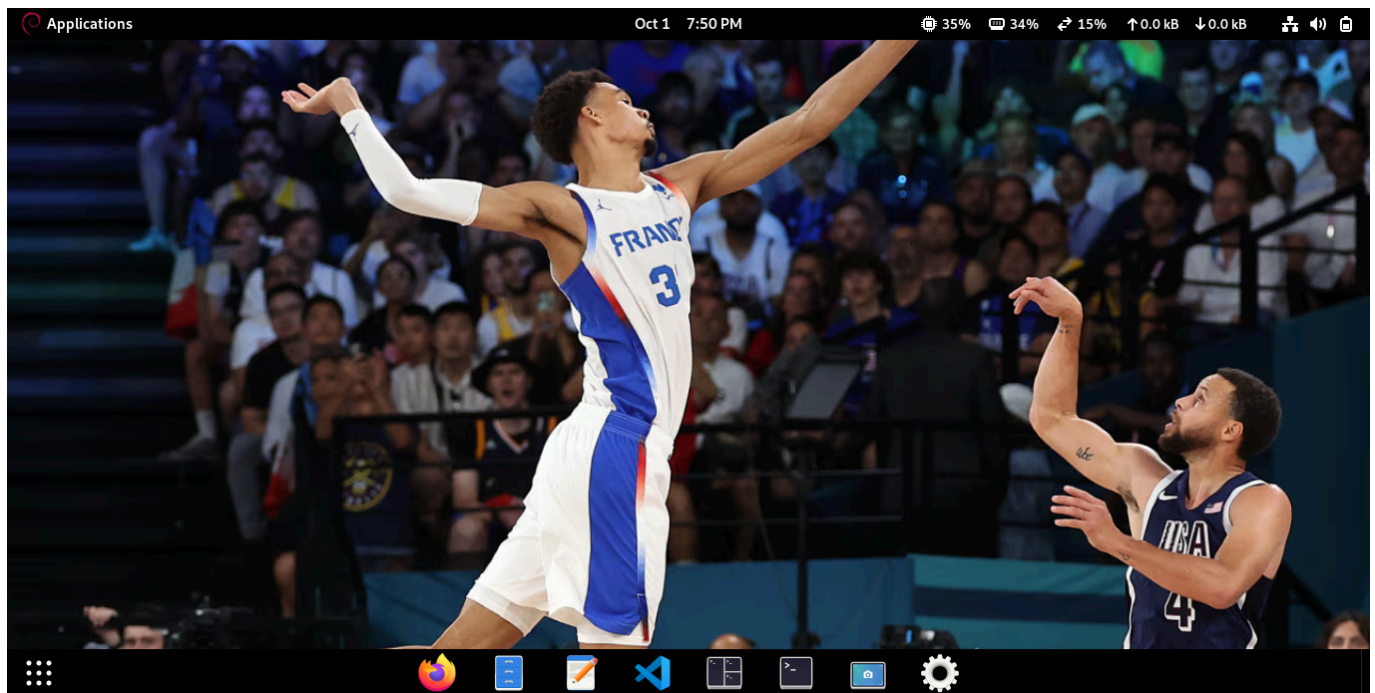
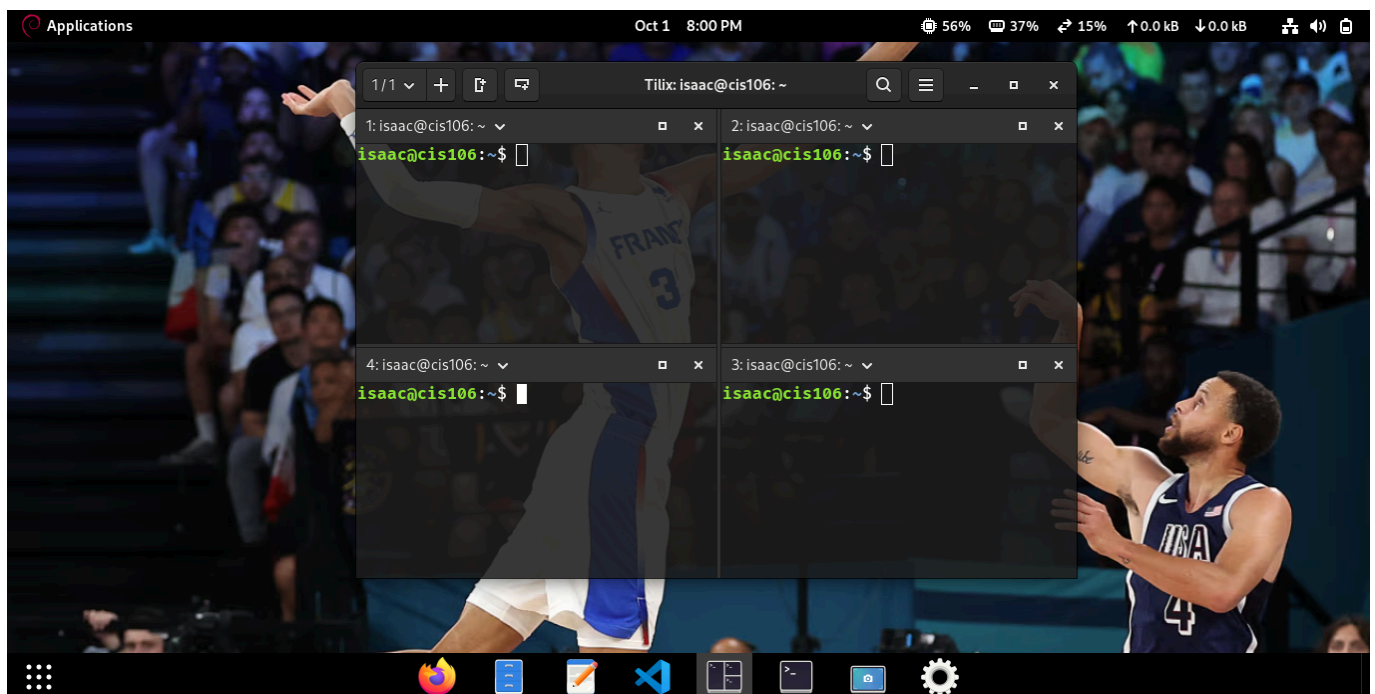


Lab 3 Submission

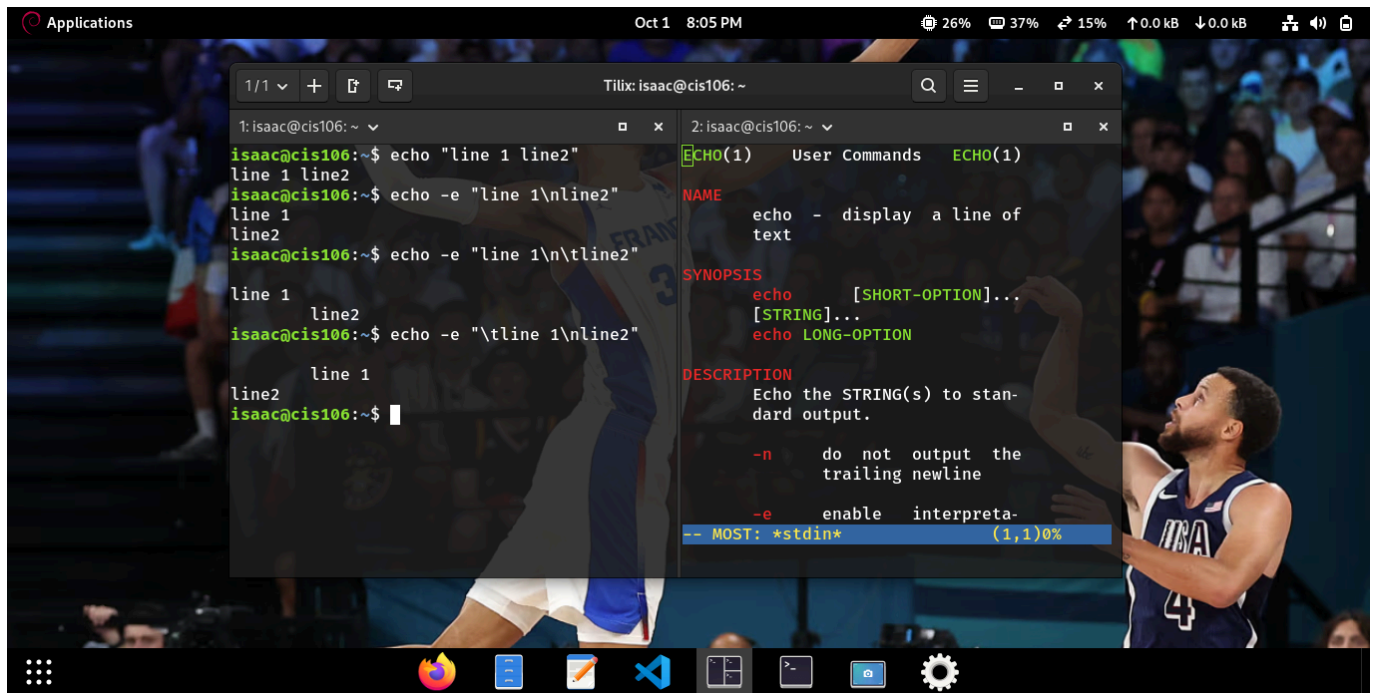
Question 1



Question 2



Question 3



The screenshot shows a Linux desktop environment. At the top, the system status bar displays 'Applications', 'Oct 1 8:05 PM', and various resource usage metrics (26% CPU, 37% memory, 15% swap, 0.0 kB network). The background is a basketball player in a blue jersey. In the foreground, a terminal window titled 'Tilix: isaac@cis106: ~' is open, showing a series of echo commands and their outputs. To the right of the terminal, a help page for the 'echo' command is displayed, detailing its usage, options, and synopsis.

```
1: isaac@cis106: ~  
isaac@cis106:~$ echo "line 1 line2"  
line 1 line2  
isaac@cis106:~$ echo -e "line 1\nline2"  
line 1  
line2  
isaac@cis106:~$ echo -e "line 1\n\tline2"  
line 1  
    line2  
isaac@cis106:~$ echo -e "\tline 1\nline2"  
    line 1  
line2  
isaac@cis106:~$
```

ECHO(1) User Commands ECHO(1)

NAME
echo - display a line of text

SYNOPSIS
echo [SHORT-OPTION]... [STRING]...
echo LONG-OPTION

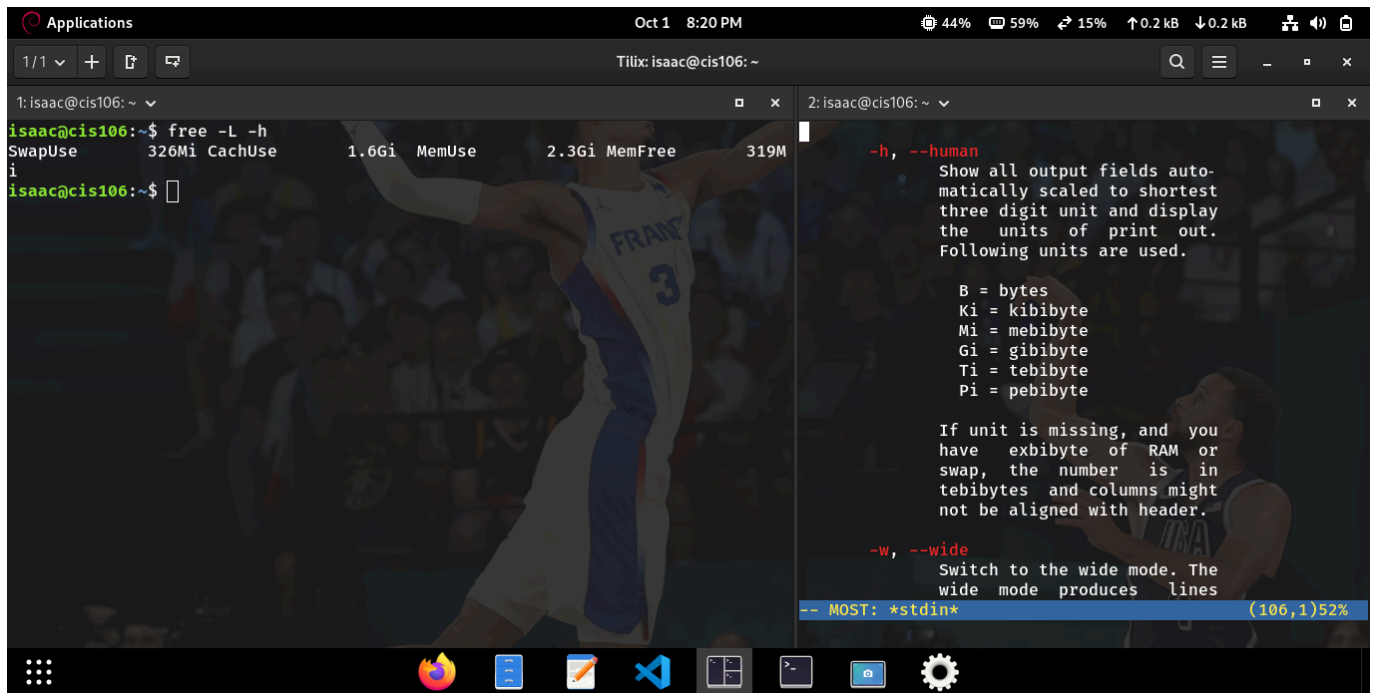
DESCRIPTION
Echo the STRING(s) to standard output.

-n do not output the trailing newline

-e enable interpretation of backslash escapes

-- MOST: *stdin* (1,1)0%

Challenge Questions



```
Applications Oct 1 8:20 PM 44% 59% 15% ↑ 0.2 kB ↓ 0.2 kB 1/1 + [search] [menu] [close]
Tilix: isaac@cis106: ~
1: isaac@cis106: ~
isaac@cis106:~$ free -L -h
              1.6Gi MemUse      2.3Gi MemFree      319M
SwapUse      326Mi CachUse
i
isaac@cis106:~$
```

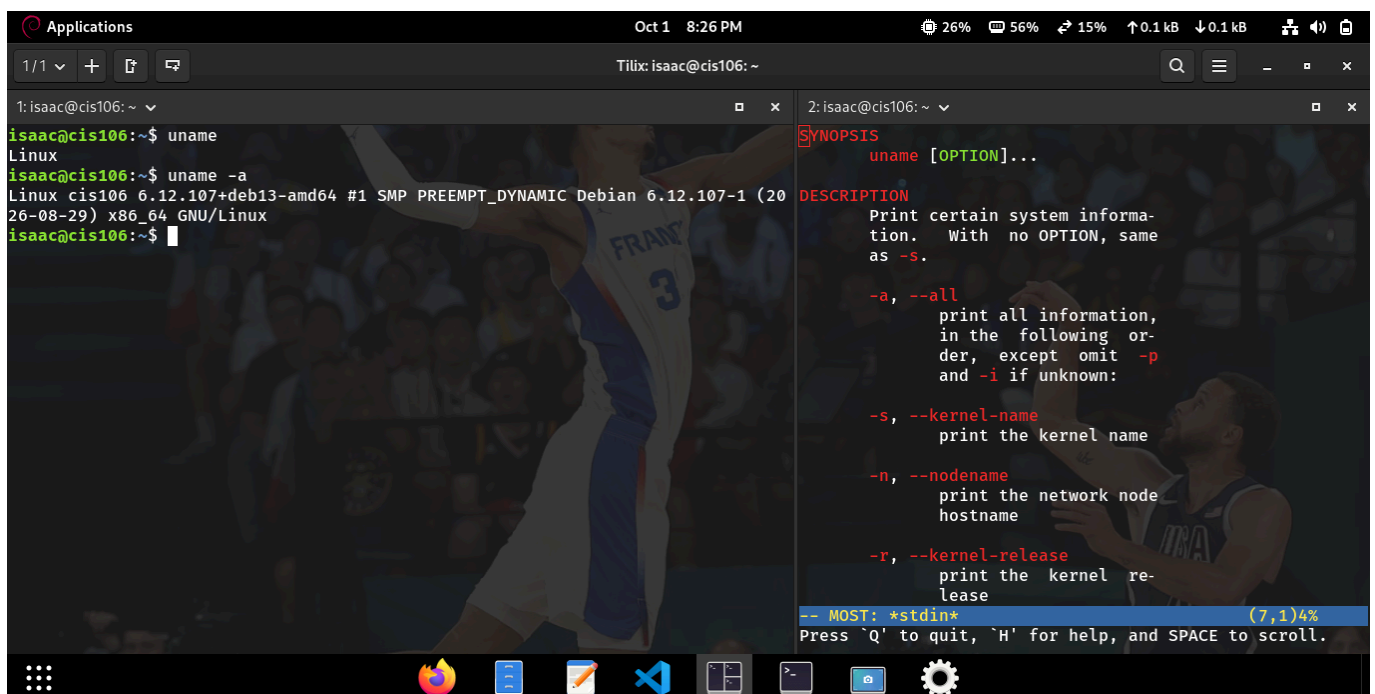
-h, --human
Show all output fields automatically scaled to shortest three digit unit and display the units of print out. Following units are used.

B = bytes
Ki = kibibyte
Mi = mebibyte
Gi = gibibyte
Ti = tebibyte
Pi = pebibyte

If unit is missing, and you have exbibyte of RAM or swap, the number is in tebibytes and columns might not be aligned with header.

-w, --wide
Switch to the wide mode. The wide mode produces lines

-- MOST: *stdin* (106,1)52%



```
Applications Oct 1 8:26 PM 26% 56% 15% ↑ 0.1 kB ↓ 0.1 kB 1/1 + [search] [menu] [close]
Tilix: isaac@cis106: ~
1: isaac@cis106: ~
isaac@cis106:~$ uname
Linux
isaac@cis106:~$ uname -a
Linux cis106 6.12.107+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.107-1 (2026-08-29) x86_64 GNU/Linux
isaac@cis106:~$
```

SYNOPSIS
uname [OPTION]...

DESCRIPTION
Print certain system information. With no OPTION, same as -s.

-a, --all
print all information, in the following order, except omit -p and -i if unknown:

-s, --kernel-name
print the kernel name

-n, --nodename
print the network node hostname

-r, --kernel-release
print the kernel release

-- MOST: *stdin* (7,1)4%
Press 'Q' to quit, 'H' for help, and SPACE to scroll.

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